

LOGISTICS ANALYTICS - DAX MEASURES

Power BI Portfolio Project | Measures Documentation

Sl.no	Measure Name	DAX	Description
1	Total Revenue	Total Revenue = SUM (Load_Summary[revenue])	Calculates total revenue generated from all loads in the current filter context.
2	Total Loads	Total Loads = DISTINCTCOUNT (Load_Summary[load_id])	Counts the number of unique loads in the current filter context.
3	Total Trips	Total Trips = DISTINCTCOUNT (Trip_Operations[trip_id])	Counts unique trips and provides the base for trip-level fleet and safety KPIs.
4	Total Service Events	Total Service Events = DISTINCTCOUNT (Delivery_Performance[event_id])	Counts unique pickup and delivery service events.
5	On-Time Service %	On-Time Service % = DIVIDE (CALCULATE ([Total Service Events], Delivery_Performance[on_time_flag] = 1), [Total Service Events], 0)	Calculates the percentage of all service events completed on time.
6	On-Time Target %	On-Time Target % = 0.95	Stores the 95% service target used as the benchmark for delivery reliability.
7	On-Time Gap %	On-Time Gap % = [On-Time Service %] - [On-Time Target %]	Calculates the percentage-point gap between actual on-time service and the 95% target.
8	Delivery On-Time %	Delivery On-Time % = DIVIDE (CALCULATE ([Total Service Events], Delivery_Performance[event_type] = "Delivery", Delivery_Performance[on_time_flag] = 1), CALCULATE ([Total Service Events], Delivery_Performance[event_type] = "Delivery"), 0)	Calculates the percentage of delivery events completed on time.
9	Delivery On-Time Gap %	Delivery On-Time Gap % = [Delivery On-Time %] - [On-Time Target %]	Shows the percentage-point gap between delivery on-time performance and the 95% target.
10	Pickup On-Time %	Pickup On-Time % = DIVIDE (CALCULATE ([Total Service Events], Delivery_Performance[event_type] = "Pickup", Delivery_Performance[on_time_flag] = 1), CALCULATE ([Total Service Events], Delivery_Performance[event_type] = "Pickup"), 0)	Calculates the percentage of pickup events completed on time.
11	Pickup On-Time Gap %	Pickup On-Time Gap % = [Pickup On-Time %] - [On-Time Target %]	Shows the percentage-point gap between pickup on-time performance and the 95% target.
12	Late Events	Late Events = CALCULATE ([Total Service Events], Delivery_Performance[on_time_flag] = 0)	Counts all pickup and delivery events that were not completed on time.
13	Late Deliveries	Late Deliveries = CALCULATE ([Total Service Events], Delivery_Performance[event_type] = "Delivery", Delivery_Performance[on_time_flag] = 0)	Counts delivery events that were completed late.
14	Delivery Late %	Delivery Late % = DIVIDE ([Late Deliveries], CALCULATE ([Total Service Events], Delivery_Performance[event_type] = "Delivery"), 0)	Calculates the percentage of delivery events that were late.

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15	Avg Delay Minutes	Avg Delay Minutes = AVERAGEX (FILTER (Delivery_Performance, Delivery_Performance[actual_datetime] > Delivery_Performance[scheduled_datetime]), DATEDIFF (Delivery_Performance[scheduled_datetime], Delivery_Performance[actual_datetime], MINUTE))	Calculates the average delay, in minutes, for events completed after their scheduled time.
16	Total Delay Minutes	Total Delay Minutes = SUMX (FILTER (Delivery_Performance, Delivery_Performance[actual_datetime] > Delivery_Performance[scheduled_datetime]), DATEDIFF (Delivery_Performance[scheduled_datetime], Delivery_Performance[actual_datetime], MINUTE))	Calculates the total number of delay minutes across late service events.
17	Total Delay Hours	Total Delay Hours = DIVIDE ([Total Delay Minutes], 60, 0)	Converts total delay minutes into hours for route-level delay analysis.
18	Total Detention Hours	Total Detention Hours = DIVIDE (SUM (Delivery_Performance[detention_minutes]), 60, 0)	Calculates total detention time in hours and is used to identify facilities creating the greatest detention burden.
19	Total Fuel Cost	Total Fuel Cost = SUM (Fuel_Analysis[total_cost])	Calculates total fuel-purchase cost in the current filter context.
20	Total Miles	Total Miles = SUM (Trip_Operations[actual_distance_miles])	Calculates the total actual miles traveled across trips.
21	Total Fuel Gallons	Total Fuel Gallons = SUM (Trip_Operations[fuel_gallons_used])	Calculates the total gallons of fuel consumed across trips.
22	Fleet MPG	Fleet MPG = DIVIDE ([Total Miles], [Total Fuel Gallons], 0)	Calculates weighted fleet fuel efficiency as total miles divided by total fuel gallons.
23	Fuel Cost per Mile	Fuel Cost per Mile = DIVIDE ([Total Fuel Cost], [Total Miles], 0)	Calculates fuel spend per actual mile traveled and helps compare truck operating efficiency.
24	Avg Idle Hours per Trip	Avg Idle Hours per Trip = DIVIDE (SUM (Trip_Operations[idle_time_hours]), [Total Trips], 0)	Calculates average idle time per trip to highlight potential fuel-efficiency opportunities.
25	Worst MPG Rank	Worst MPG Rank = RANKX (ALL (Dim_Trucks[unit_number]), [Fleet MPG], , ASC, DENSE)	Ranks trucks from lowest to highest MPG. It is used to filter the Bottom 10 Trucks by Fuel Efficiency visual.
26	Total Maintenance Cost	Total Maintenance Cost = SUM (Maintenance_Analysis[total_cost])	Calculates total maintenance expenditure.
27	Maintenance Events	Maintenance Events = DISTINCTCOUNT (Maintenance_Analysis[maintenance_id])	Counts unique maintenance service events.
28	Total Downtime Hours	Total Downtime Hours = SUM (Maintenance_Analysis[downtime_hours])	Calculates total vehicle downtime caused by maintenance events.

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29	Avg Downtime per Event	Avg Downtime per Event = DIVIDE ([Total Downtime Hours], [Maintenance Events], 0)	Calculates average downtime hours generated by each maintenance event.
30	Total Parts Cost	Total Parts Cost = SUM (Maintenance_Analysis[parts_cost])	Calculates total cost of replacement parts used in maintenance.
31	Total Labor Cost	Total Labor Cost = SUM (Maintenance_Analysis[labor_cost])	Calculates total labor expenditure associated with maintenance work.
32	Safety Incidents	Safety Incidents = DISTINCTCOUNT (Safety_Analysis[incident_id])	Counts unique safety incidents recorded in the current filter context.
33	Preventable Incidents	Preventable Incidents = CALCULATE ([Safety Incidents], Safety_Analysis[preventable_flag] = 1)	Counts safety incidents classified as preventable.
34	Preventable Incident %	Preventable Incident % = DIVIDE ([Preventable Incidents], [Safety Incidents], 0)	Calculates the percentage of recorded safety incidents that were preventable.
35	Injury Incidents	Injury Incidents = CALCULATE ([Safety Incidents], Safety_Analysis[injury_flag] = 1)	Counts safety incidents that involved an injury.
36	At-Fault Incidents	At-Fault Incidents = CALCULATE ([Safety Incidents], Safety_Analysis[at_fault_flag] = 1)	Counts incidents where the company driver was marked as at fault.
37	Total Claim Amount	Total Claim Amount = SUM (Safety_Analysis[claim_amount])	Calculates total claim value associated with safety incidents.
38	Total Damage Cost	Total Damage Cost = SUM (Safety_Analysis[vehicle_damage_cost]) + SUM (Safety_Analysis[cargo_damage_cost])	Calculates combined vehicle and cargo damage cost from safety incidents.
39	Incidents per 1000 Trips	Incidents per 1000 Trips = DIVIDE ([Safety Incidents], [Total Trips], 0) * 1000	Normalizes safety incidents by trip volume, showing the number of incidents per 1,000 trips.
40	Total Operating Cost	Total Operating Cost = [Total Fuel Cost] + [Total Maintenance Cost] + [Total Claim Amount]	Combines fuel, maintenance and claim costs into the operating-cost figure used on the Executive Overview.
41	Fuel Cost %	Fuel Cost % = DIVIDE ([Total Fuel Cost], [Total Operating Cost], 0)	Calculates fuel's share of the analyzed operating cost.
42	Maintenance Cost %	Maintenance Cost % = DIVIDE ([Total Maintenance Cost], [Total Operating Cost], 0)	Calculates maintenance's share of the analyzed operating cost.
43	Claim Cost %	Claim Cost % = DIVIDE ([Total Claim Amount], [Total Operating Cost], 0)	Calculates safety claims' share of the analyzed operating cost.
44	Data As Of	Data As Of = CALCULATE (MAX ('Dim_Date'[Date]), REMOVEFILTERS ('Dim_Date'))	Returns the latest date in the model regardless of Year or Month slicers. Format the measure as dd MMM yyyy.

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45	Revenue YoY % (2024 vs 2023)	Revenue YoY % (2024 vs 2023) = VAR Revenue2024 = CALCULATE ([Total Revenue], REMOVEFILTERS ('Dim_Date'), 'Dim_Date'[Year] = 2024) VAR Revenue2023 = CALCULATE ([Total Revenue], REMOVEFILTERS ('Dim_Date'), 'Dim_Date'[Year] = 2023) RETURN DIVIDE (Revenue2024 - Revenue2023, Revenue2023, 0)	Calculates complete-year revenue change for 2024 versus 2023, avoiding the incomplete January 2025 period.
46	Fleet MPG YoY % (2024 vs 2023)	Fleet MPG YoY % (2024 vs 2023) = VAR MPG2024 = CALCULATE ([Fleet MPG], REMOVEFILTERS ('Dim_Date'), 'Dim_Date'[Year] = 2024) VAR MPG2023 = CALCULATE ([Fleet MPG], REMOVEFILTERS ('Dim_Date'), 'Dim_Date'[Year] = 2023) RETURN DIVIDE (MPG2024 - MPG2023, MPG2023, 0)	Calculates complete-year fleet MPG change for 2024 versus 2023.
47	Fuel Cost YoY % (2024 vs 2023)	Fuel Cost YoY % (2024 vs 2023) = VAR Fuel2024 = CALCULATE ([Total Fuel Cost], REMOVEFILTERS ('Dim_Date'), 'Dim_Date'[Year] = 2024) VAR Fuel2023 = CALCULATE ([Total Fuel Cost], REMOVEFILTERS ('Dim_Date'), 'Dim_Date'[Year] = 2023) RETURN DIVIDE (Fuel2024 - Fuel2023, Fuel2023, 0)	Calculates complete-year fuel-cost change for 2024 versus 2023. A negative result means fuel cost decreased.
48	Maintenance Cost YoY % (2024 vs 2023)	Maintenance Cost YoY % (2024 vs 2023) = VAR Maintenance2024 = CALCULATE ([Total Maintenance Cost], REMOVEFILTERS ('Dim_Date'), 'Dim_Date'[Year] = 2024) VAR Maintenance2023 = CALCULATE ([Total Maintenance Cost], REMOVEFILTERS ('Dim_Date'), 'Dim_Date'[Year] = 2023) RETURN DIVIDE (Maintenance2024 - Maintenance2023, Maintenance2023, 0)	Calculates complete-year maintenance-cost change for 2024 versus 2023. A negative result means maintenance cost decreased.
49	Safety Incidents YoY % (2024 vs 2023)	Safety Incidents YoY % (2024 vs 2023) = VAR Incidents2024 = CALCULATE ([Safety Incidents], REMOVEFILTERS ('Dim_Date'), 'Dim_Date'[Year] = 2024) VAR Incidents2023 = CALCULATE ([Safety Incidents], REMOVEFILTERS ('Dim_Date'), 'Dim_Date'[Year] = 2023) RETURN DIVIDE (Incidents2024 - Incidents2023, Incidents2023, 0)	Calculates complete-year safety-incident change for 2024 versus 2023. A positive result indicates more incidents.